

Benjamin (Benji) Adjepong

U.S. CITIZEN, ELIGIBLE TO OBTAIN CLEARANCE

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OBJECTIVE

Seeking experiential learning opportunities such as internships/co-ops to apply technical skills in a collaborative environment

EDUCATION

B.S. COMPUTER ENGINEERING - VIRGINIA TECH | BLACKSBURG, VA

GPA: 3.55

EXPECTED SPRING 2025

Major: Machine Learning

Secondary Focus: Controls, Robotics & Autonomy (CRA)

Minor: Computer Science

RELEVANT COURSEWORK: Data Structures & Algorithms (C++), Embedded Systems, AI & Engineering Applications

EXPERIENCE

MICROSOFT | REDMOND, WA

MAY 2024 – AUGUST 2024

Software/Firmware Engineer Intern – Surface Development Team

- Spearheaded the design and implementation of low-level drivers, tools, and firmware within embedded systems
- Conducted thorough testing/refining to ensure products met high standards for reliability, efficiency, and user experience
- Collaborated closely with cross-functional peers in a dynamic and multidisciplinary environment

COLLINS AEROSPACE | STERLING, VA

JUNE 2023 – AUGUST 2023

SEPP Mission Systems Software Engineer Intern – Station Keeping Playbook (SKP)

- Developed operational testing platform for unmanned aerial systems (UAS) on ground robots reducing costs by up to 90%
- Tested autonomous capabilities for UAS's by integrating autonomous flight software onto ground robots using Py-C bindings
- Learned and implemented various python libraries to process GPS & IMU data to command motors based on SKP outputs
- Presented project results to company leadership at HQ through PowerPoint presentations and a booth setup at the SEPP fair

VIRGINIA TECH NATIONAL SECURITY INSTITUTE | BLACKSBURG, VA

SEPTEMBER 2023 – PRESENT

Project Manager – Edge Supercomputing for Autonomous Sensor Platforms

- Learned and applied the basics of program management under government projects
- Composed project budgets and schedules, and reported on performance risks/challenges

HUME CENTER FOR NATIONAL SECURITY & TECHNOLOGY | BLACKSBURG, VA

DECEMBER 2022 – MAY 2023

Paid Undergraduate Research Assistant – Boeing IoT Device Characterization

- Conducted extensive experiments testing range, battery and durability of Bluetooth Low Energy (BLE) IoT devices
- Analyzed sensor data using Putty and Python scripting to identify behavioral trends
- Presented detailed reports on the findings of the experiments to IoT development team at Boeing

RELEVANT PROJECTS

AUTONOMOUS TRACTOR | ARDUINO, PYTHON

SPRING 2023

- Developed graphical user interface (GUI) for an autonomous tractor using Python and the PyGame library
- Collaborated with partner to interface Arduino Uno with 4+ sensors, enabling the tractor to traverse a course autonomously

LOW-POWER TAMAGOCHI | EMBEDDED C

SPRING 2023

- Created an interactive virtual pet minigame on TI MSP432 microcontroller board themed after *Amazing World of Gumball*
- Configured several hardware abstraction layers for input and output for joystick, buttons and LEDs

SKILLS

LANGUAGES

- Embedded C, C, C++, Python, Rust, R, Verilog/VHDL

TECHNICAL TOOLS

- Arduino, Nvidia Jetson Nano, ROSMASTER X3, TI MSP432 Launchpad & Boosterpack, AD2 Oscilloscope, DE10-Lite FPGA

SOFT SKILLS

- Effective Communication, Adaptability, Time Management

EXTRACURRICULAR ACTIVITIES

NATIONAL SOCIETY OF BLACK ENGINEERS | VT CHAPTER SENATOR

SEPTEMBER 2021 – PRESENT

- Organized and coordinated 5+ social and academic events for the general student body at Virginia Tech
- Participated in 10+ professional development events such conferences and resume review workshops

VIRGINIA TECH COMPETITIVE ROBOT ORGANIZATION | A-TEAM ENGINEER

AUGUST 2023 – PRESENT

- Rapidly prototyped to create an autonomous robot to optimally complete tasks in a simple and reliable manner
- Developed and implemented line following algorithm enabling robot to autonomously traverse the competition course

- **INTERESTS:** Embedded Software Systems, Robotics, Machine Learning

- **HOBBIES:** E-skating, Fitness, Ping-Pong, Basketball, Soccer



LinkedIn